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**A Major Project Report**
**(Phase-I)**

**On**

# Intelligent Farm Resource Optimization Using ML and Predictive Analytics
# 26P_3CSE139

***Submitted in partial fulfillment of the requirement for the award of the degree of***

## BACHELOR OF TECHNOLOGY

IILM UNIVERSITY

Session 2026-27

in

Computer Science and Engineering

By


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Under the guidance of **Dr. SWATI VASHISHT**

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
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May, 2026



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## CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**Intelligent Farm Resource Optimization Using ML and Predictive Analytics** ", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Dr. SWATI VASHISHT**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Signature of Guide Name and Details of Guide

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# 1. ABSTRACT

The rapid growth of the global population has increased the demand for agricultural productivity while placing significant pressure on limited resources such as water, fertilizers, energy, and labor. Traditional farming practices, particularly in small- and medium-scale agricultural operations, often rely on fixed irrigation schedules, visual inspection, regional guidelines, and manual decision-making rather than real-time information about soil conditions, crop health, and weather patterns. This approach can result in inefficient resource utilization, reduced crop productivity, and environmental damage.

This project proposes an **Intelligent Precision Agriculture System** that uses real-time data and smart monitoring technologies to enable efficient, data-driven, and preventive crop management. The system integrates soil-moisture sensing, environmental monitoring, weather information, and crop-condition analysis to determine the actual requirements of crops. Based on these parameters, irrigation and other agricultural inputs can be applied according to crop needs rather than predetermined schedules. The system can also identify variations in crop growth zones, allowing fertilizers and pesticides to be targeted more precisely and reducing unnecessary application.

By continuously collecting and analyzing field data, the proposed system aims to shift agricultural management from a reactive approach to a **preventive and predictive approach**. Early identification of water stress, unfavorable environmental conditions, and potential pest or crop-health problems can help farmers take timely corrective actions. The overall objective is to improve crop yield and resource efficiency while reducing water consumption, excessive chemical usage, environmental pollution, and dependence on manual crop monitoring. The proposed solution therefore provides a scalable and sustainable approach toward modern precision farming and supports the long-term goal of achieving higher agricultural productivity with fewer resources.

## 2.Problem Definition

Meeting the demands of a growing population becomes increasingly difficult, where, agriculture continues to face issues regarding the amounts of available resources. Water tables continue to drop in most farming regions while the costs of fertilizer and energy continue to increase; moreover, manual labor for crop monitoring is diminishing. However, the need for higher crop yields still persists.

Due to various factors, most small to medium sized farming operations continue to water, fertilize, use pesticides, and provide labor to their crops according to schedules, visual estimates, or rules of thumb for their region, rather than with real-time data on soil, crops, and the weather.

Over time, these mismatches lead to three problems:

- Inefficient water usage, where irrigation is scheduled according to a time frame calendar and not soil moisture levels, evapotranspiration, or short-term forecasts.
- Increase in environmental pollution due to excess fertilizer and pesticide used because it is applied evenly across a field as opposed to being directed to specific crop growth zones.
- After a crop has been stressed or pests have caused damage, farmers use multiple sensors and trend data in a reactive manner as opposed to preventive farming.

## 3.Literature Survey

Four main areas of research are related to this problem. These overlap as follows:

- Precision irrigation and soil moisture prediction. With historical soil sensor readings and weather forecasts, time-series and LSTM-based models allow farmers to predict soil moisture and irrigation requirements several days in advance with a water reduction of up to 40% after irrigation scheduling. Some of the existing approaches are based on reinforcement learning, which allows the irrigation valves to be set based on feedback instead of rigid rules.
- Remote sensing and crop health monitoring. Before they become visible to the human eye, CNNs (convolutional neural networks) applied to multispectral images (NDVI-Normalized Difference

Vegetation Index) have been able to detect stress, nutrient deficiencies, and pests and diseases in the crop. The existing literature requires either a high-cost imaging system or is somewhat limited due to the intervals of satellite revisits.

- Yield prediction and resource forecasting. Forecasting end of season yields combined with historical yields of soil, and climatic data together with ensemble methods have also been used to determine the optimal fertilizer application. This is mostly accurate at the regional level, but not at the farm/field level due to poor data available for a specific location.
- IOT-based farm monitoring platforms. Although several monitoring platforms (commercial and open source) collect soil, weather, and other data on a farm, most focus on real time data collection and on-farm monitoring.

**Gap identified:** Many existing models target a single component of yield prediction, pest detection, or irrigation. The few multi-component systems range from prohibitively expensive to the farmer, to insightfully designed and yet in the form of view-only dashboards. Farmers require a unified optimization plan that considers water, fertilizer, and labor together to improve crop yield, in a predictive pipeline that integrates soil, weather, imagery, and historical yield data.

## 4.Objectives

The first objective is to create a data pipeline that consolidates and cleans multi-source data for each field zone. This includes data from: soil sensors (moisture, pH, NPK), data from weather stations/APIs, and data from crop imagery.

The second objective is to create machine learning models. These models should be able to predict: short-term irrigation need, the fertilizer/nutrient need for each zone, and the potential yield for each zone, given the data and the forecasted conditions.

The third objective is to create a recommendation system that transforms the predictions created by the models into a optimization based plan that is the easiest to carry out, for example, "irrigate zone three for 20 minutes tomorrow with a hold on the application of fertilizer in zone one."

The fourth objective is to evaluate system resources. This includes how much water and fertilizer is used and how accurate the predicted yield is.

This system should be measured against a control system of fixed time intervals.

The last objective is to develop an application for the system with minimal technical knowledge needed to use it.

## 5. Scope

In scope:

- Collection of soil moisture, temperature, humidity, rainfall, and basic soil nutrient data for a defined pilot area (single farm or cluster of nearby fields).
- ML models for scheduling irrigation, recommending fertilizer, and predicting yield with historical and/or publicly available agricultural data as well as the pilot sensor data.
- An automatic system to recommend actions to farmers based on predictions of a rule/ML algorithm.
- An app or dashboard for current and forecast conditions and recommendations.
- Evaluating the impact of using a recommendation system for irrigation scheduling with a comparison of baseline (fixed-schedule) irrigation.

## 6. Feasibility Study

- **Technical feasibility**: The main ML techniques for the project (time-series forecasting, regression, ensemble models, and recommendation logic) exist and have robust libraries (scikit-learn, TensorFlow/PyTorch, pandas). Cheap capacitive soil sensors connected to microcontrollers are available to measure soil moisture and can be supplemented with sensors to measure temperature and humidity. Weather forecast data can be obtained from free public APIs, so a private weather station is not needed for the pilot phase.

- **Data feasibility**: There are pre-existing public datasets of agriculture and climate that can be used to train and validate models and alleviate the cold-start problem, even prior to the availability of pilot

sensors.

- **Economic feasibility**: A low-cost deployment using a few nodes and an edge (e.g. Raspberry Pi) computing unit is possible with a low budget compared to the high costs assumed in competing commercial precision agriculture systems.
- **Operational feasibility**: The end user of the system builds no ML models. Recommendations are delivered as prioritized actions to the end user.
- **Schedule / Resources**: The deliverables for the project (a data pipeline, three models, a recommendation engine, and a dashboard) are in scope and can be completed in the time available for an academic project and thesis.

## 5. References

[1] “Machine Learning Based Soil Moisture Prediction for Internet of Things Based Smart Irrigation System,” IEEE Conference Publication, 2019.

[2] “Predictive Water Management for Precision Farming Using IoT & Bi-LSTM Networks,” IEEE Conference Publication.

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[6] “Smart Agriculture Monitoring System Using Internet of Things (IoT),” IEEE Conference Publication.

# 5. PLAGIARISM REPORT

